CLAIMS

What is claimed is:

1. A covert illumination and imaging apparatus, comprising:

    - a translucent enclosure shaped as a standard illumination bulb;
    - a curved active-matrix substrate conforming to at least a portion of an interior surface of said enclosure;
    - an interleaved array of light-emitting subpixels and light-receiving photodetector subpixels fabricated on said curved active-matrix substrate; and
    - a driver circuit configured to execute a de-phased duty cycle, wherein said light-emitting subpixels are driven to emit visible illumination into a surrounding environment, and wherein said photodetector subpixels are selectively activated to capture ambient light during intermittent dark intervals when said light-emitting subpixels are deactivated.

2. The apparatus of claim 1, wherein the light-emitting subpixels comprise organic light-emitting diodes (OLED) or quantum dot electroluminescent (QDEL) diodes.

3. The apparatus of claim 1, wherein the photodetector subpixels comprise organic photodetectors (OPD).

4. The apparatus of claim 1, wherein the total light output of the light-emitting subpixels is configured to match the standardized lumen output of a conventional single-purpose light bulb.

5. The apparatus of claim 1, wherein the curved active-matrix substrate provides a panoramic or substantially hemispherical field of view for the photodetector subpixels.

6. The apparatus of claim 1, further comprising a computational processor housed within a base of the apparatus, said processor configured to reconstruct an image of the surrounding environment from voltage signals generated by the photodetector subpixels.